

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

M. Tech. Electrical Semester – II, Examination, December 2013

EE 707 Advanced Power System Dynamics and Stability

Date: 24/12/2013, Tuesday

Time: 10.00 a.m. to 01.00 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each section.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary

SECTION – I

- Q – 1 A. Give the classification and definition of power system stability and explain each in briefly 07

OR

- A. Explain the procedure to draw PV and QV curve for voltage stability 07
- Q – 2 A. List out the different methods for voltage stability analysis. Explain sensitivity analysis and modal analysis method to predict the voltage stability. 07
- B. Derive the mathematical modeling of separately excited DC exciter. 07

OR

- B. Explain any one type of excitation system with schematic 07
- Q – 3 **Attempt any two** 14
- (a) Power system stabilizer for small signal stability enhancement
- (b) For two bus system, explain how SVC can increase the stability margin?
- (c) Explain different methods for improvement in transient stability

SECTION – II

- Q – 4 A. Derive the block diagram of SMIB system with classical model of generator 07

OR

- A. Discuss the effect of field circuit dynamics on small signal stability of SMIB system 07
- Q – 5 A. Derive the shaft system torque equation for sub-synchronous oscillations 07
- B. Explain the following term with respect to sub-synchronous oscillations 07
- (1) Shaft system model (2) Torsional stiffness and (3) Damping coefficient

OR

- B. Explain the phenomena of SSR in detail considering series compensated 07
transmission network and self excitation due to induction generator effect

Q – 6

Attempt Any Two

14

- (a) Derive the mathematical model for following
(i) AC exciter and rectifiers (b) excitation system stabilizing circuit
(b) Write a short note on SSSC for stability enhancement
(c) With schematic, explain the principle of TCSC and give the advantages of TCSC
